

Classwork 2

Exponentials and logarithms

Grades: [A] [M] [E]

Answers · tutor's working, not an official mark scheme

Show all working. Give answers to 3 decimal places unless the question asks for an exact answer.

Part A · Switch, combine, solve

A1 Evaluate $\log_3 243$. [A]

$$3^5 = 243, \text{ so } \log_3 243 = 5$$

A2 Solve $\log_6 x = 2$. [A]

$$x = 6^2 = 36$$

A3 Write $3 \log 2 + \log 5 - \log 4$ as a single logarithm, and find its value. [A]

$$\log 8 + \log 5 - \log 4 = \log \frac{8 \times 5}{4} = \log 10 = 1$$

A4 Solve $6^x = 75$. [A]

$$x \log 6 = \log 75, \quad x = \frac{\log 75}{\log 6} = 2.410$$

A5 Solve $\log_x 16 = 4$. [A]

$$x^4 = 16, \text{ so } x = 2 \text{ (the base cannot be } -2)$$

Part B · Two logs, and a model

B1 Solve $2^{3x-2} = 50$. [M]

$$(3x - 2) \log 2 = \log 50, \quad 3x - 2 = \frac{\log 50}{\log 2} = 5.6439$$

$$3x = 7.6439, \quad x = 2.548$$

B2 Solve $\log_5(x+1) + \log_5(x-3) = 1$. [M]

$$\log_5((x+1)(x-3)) = 1, \quad (x+1)(x-3) = 5, \quad x^2 - 2x - 8 = 0$$

$$(x-4)(x+2) = 0: \quad x = 4 \quad (x = -2 \text{ rejected: } \log_5(-1) \text{ does not exist})$$

B3 A phone bought for \$1200 loses 18% of its value each year. When is it first worth less than \$300? [M]

$$1200(0.82)^t = 300, \quad 0.82^t = 0.25$$

$$t = \frac{\log 0.25}{\log 0.82} = 6.99$$

It is first worth less than \$300 during the 7th year, after about 7.0 years.

Part C · A quadratic in disguise

C Solve $4^x - 5 \times 2^x + 4 = 0$. Explain each step.

[E]

Words you may use

let $u = 2^x$ · $4^x = (2^x)^2$ · factorise · change back · always positive

Sentence starters (optional): *Since* $4 = 2^2$, ... *Letting* $u = 2^x$, ... *So* $2^x = \dots$

Since $4^x = (2^2)^x = (2^x)^2$, let $u = 2^x$: $u^2 - 5u + 4 = 0$ MP1

$(u - 1)(u - 4) = 0$, so $u = 1$ or $u = 4$ MP2

$2^x = 1$ gives $x = 0$; $2^x = 4$ gives $x = 2$ MP3

Both values of u are positive, so neither is rejected (2^x is always positive).

MP1 and MP2 are r; both values of x with the reasoning is t.

Extension

Solve $2^x = 3^{x-1}$.

$x \log 2 = (x - 1) \log 3$, $x(\log 3 - \log 2) = \log 3$

$x = \frac{\log 3}{\log 3 - \log 2} = 2.710$